

Design of Combinational Logic Circuits using Simulated Annealing

Pavitra Y J
Department of Electronics &
Communication Engineering
PES University
Bangalore, India
pavitra.yj@pes.edu

Jamuna S
Department of Electronics &
Communication Engineering
Dayananda Sagar College of
Engineering
Bangalore, India
jamuna-ece@dayanandasagar.edu

Manikandan J
Department of Electronics &
Communication Engineering
PES University
Bangalore, India
manikandanj@pes.edu

Arun E
Department of Electronics &
Communication Engineering
PES University
Bangalore, India
edupugantiarun@gmail.com

Abstract— Conventional methods to design combinational logic circuits (CLCs) is time consuming and needs expert knowledge. Evolutionary computing techniques have proved to be a competitive field for the evolution of CLCs. Simulated annealing is a metaheuristic which helps in finding a global optimum for a given function. The proposed work aims to design CLCs using simulated annealing (SA). Various circuits proposed in the literature are realized and experiments reveal that a maximum of 33.33% of resources are saved and 2.0x speed enhancement is achieved over the circuits reported in literature. The proposed work acquires the design requirements from the designer/user to yield scripts for FPGA implementation.

Keywords— *Combinational logic circuits, Fitness function, Optimization, Simulated Annealing*

I. INTRODUCTION

Evolutionary algorithms are metaheuristics which resemble the natural reproduction mechanism to solve complex problems. These algorithms are inspired by the biological processes like recombination, mutation and selection. Simulated annealing is a metaheuristic that performs better for discrete problems over continuous problems.

SA algorithm overcomes from being trapped in local optima and has proved to be effective for a variety of applications, such as scheduling and sequencing in [1], to find pareto solutions for electromagnetic devices and reduce CPU time in [2], to design optimal decoupling network for minimizing power density network noise in [3]. One of the drawbacks of simulated annealing is its initial sample selection, which affects the convergence of algorithm. To overcome this limitation, population based simulated annealing is proposed in [4] to find nodes with maximal node connectivity and to minimize the expected cost of repair for single failure trouble shooting model in [5].

The purpose of optimization is to achieve the “best” design based on a set of predefined criteria or constraints. The maximization/minimization of an objective function depends on the application chosen like manufacturing industry, design industry, software development, bio-medical, healthcare etc. Optimization has been employed for design of digital circuits

too. Traditional methods like Boolean algebra, Karnaugh map, Quine McCluskey and Shannon’s method are restricted to lower order circuits and it becomes cumbersome for higher order circuits. Evolutionary algorithms have been successfully employed for the realization of lower order digital circuits using various resource types. A review on evolutionary algorithms for resource optimization is reported in [6]. Optimization of gate count using population based genetic algorithm is proposed in [7], Multilayer Chromosome Evolutionary algorithm (MLCEA) for gate count optimization using modified Roulette selection in [8], extended to transistor count optimization using MLCEA and fan selection in [9], gate count optimization using genetic algorithm(GA) and Elitist selection method in [10], transistor count optimization using Imperialist Competitive algorithm(ICA) in [11], fitness function for GA to realize and optimize digital circuit with lesser gate count and generations in [12], Cluster based GA in [13], MUX count optimization using genetic algorithm in [14], extended to Reed Muller(RM) block optimization in [15,16], gate count optimization using genetic programming in [17], improvement in number of generations, elapsed time, minimization of number of gates through new constraint fitness function for Genetic Programming in [18] and optimization of gate count and transistor count using Evolutionary algorithm without crossover (EAWC) in [19]. In this paper, an attempt is made to design and evaluate the realization of CLCs using population based simulated annealing. A computer aided software is designed using C++ programming language to generate HDL scripts and test benches for design under test.

The organization of paper is as follows. Section 2 gives an introduction to simulated annealing and realization of CLC using simulated annealing is explained in section 3. Section 4 discusses the experimental results conclusion in section 5 followed by references.

II. INTRODUCTION TO SIMULATED ANNEALING

Annealing is a process in metallurgy where metals are slowly cooled to make them reach a state of low energy where they are very strong. Simulated annealing is an analogous method for optimization, typically described in

terms of thermodynamics. Simulated annealing is a process where the temperature is reduced slowly, starting from a random search at high temperature, eventually approaching zero temperature. The random movement corresponds to high temperature and there is little randomness at low temperature. The randomness should tend to jump out of local minima and find regions that have a low heuristic value. Simulated annealing algorithm is excellent at avoiding numerous local optimum and is much better on average at finding an approximate global optimum.

Population based SA commences with the random selection of m initial samples, given as

$$X = \{x_1, x_2, \dots, x_m\} \quad (1)$$

where x_i represents a randomly generated circuit of fixed length with r predefined resources as shown in Fig. 1. The predefined resources are listed by the user and it may consist of either a single type of resource or a combination of multiple resources. An illustration of sample x_i represented using combination of logic gates and 2:1 MUX is given in Fig. 1(a) and 1(b) respectively. In Fig.1, first row lists the type of resource and second row lists the inputs and outputs (O , O^i) of these resources. The possible inputs for these resources include logic 0(F), logic 1(T), primary inputs or output from previous stage (O^i).

Type	AND			OR			XOR			XNOR		
Inputs	a	b	O	b	O^i	O	0	a	O	a	1	O

(a) Using logic gates

Type	MUX				MUX				MUX				MUX			
Inputs	0	1	a	O	a	0	b	O	b	a	1	O^i	b	a	0	O

(b) Using 2:1 MUX

Fig. 1. Representation of randomly generated CLC

SA requires an annealing schedule, which specifies how the temperature is reduced as the search progresses. For the proposed work, each x_i is assigned with an initial temperature of $T_i=20$.

Simulated annealing comprises of two stages: perturbation and evaluation. Perturbation is analogous to mutation in genetic algorithm. Perturbation of sample x_i given in Fig. 1(a) and 1(b) is illustrated in Fig. 2(a) and 2(b) respectively with the perturbed resource type or input highlighted in yellow color.

Type	AND			OR			NOT			XNOR		
Inputs	a	1	O	b	O^i	O	0	-	O	b	1	O

(a) For circuits using logic gates

Type	MUX				MUX				MUX				MUX			
Inputs	0	b	a	O	a	0	b	O^i	b	a	1	O^i	b	1	0	O

(b) For circuits using 2:1 MUX

Fig. 2. Representation of CLC after perturbation

The perturbed population obtained for initial population X is given as

$$X_p = \{x_{1p}, x_{2p}, \dots, x_{mp}\} \quad (2)$$

The evaluation stage of simulated annealing includes the computation of fitness score (FS) for every individual sample x_i in X and x_{ip} in X_p using (3)

$$FS = 2^n - \sum_{k=0}^{2^n-1} (Z_i \oplus Y_i) \quad (3)$$

where n denotes the number of primary inputs, Z_i is output obtained by the circuit and Y_i is the truth table output. Let FS_{old} and FS_{new} represent the fitness score obtained for X and X_p respectively. The sample x_i in initial population is replaced with perturbed sample x_{ip} if $FS_{new} > FS_{old}$. For samples with $FS_{new} \leq FS_{old}$, the condition to replace initial sample with perturbed sample is given as

$$e^{(FS_{new}-FS_{old})/T} \geq s \quad (4)$$

where T is the present temperature of the sample and s is a random value generated between 0 and 1.

The above mentioned process of perturbation and evaluation of FS is repeated for predefined G_I generations. At the end of G_I generations, the temperature of sample with best score is reset to 0. The cooling process is initiated for rest of the samples and their temperature is reduced by preset cooling factor. The process is repeated for maximum G generations and the best solution (circuit) obtained at the end of G generations is the optimal solution of interest.

III. REALIZATION OF COMBINATIONAL LOGIC CIRCUITS USING SIMULATED ANNEALING

Consider a full adder circuit with inputs a , b , c and outputs sum and carry. The best solution obtained for the realization of full adder using proposed work is shown in Fig. 3. The predefined resource count was set to 10 and the same is inferred from Fig. 3 as index 0 to 9. It is observed from Fig. 3 that two-input AND, OR, XOR gates along with wires are used for the realization of full adder. In Fig. 3, black solid arrow lines represent the primary inputs to a resource, dotted lines represent the output from a resource and blue solid arrow lines represent the final output of the circuit. Resources with no contribution in the realization of final circuit are marked with blue colored $\times$ in Fig. 3.

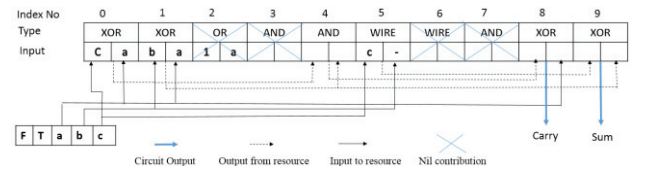


Fig. 3. Illustration for realization of Full Adder circuit using Simulated Annealing

The snapshot of Graphical User Interface (GUI) and output screen is shown in Fig. 4 for a full adder circuit, which provides information about resources used, connection between resources, truth table for the circuit generated, circuit diagram, HDL script and its test bench.

The proposed work is capable of realizing digital circuits with optimum gate count, transistor count or levels based on design requirements fed by user. Optimum gate count refers to minimum number of gates required to realize a circuit. It

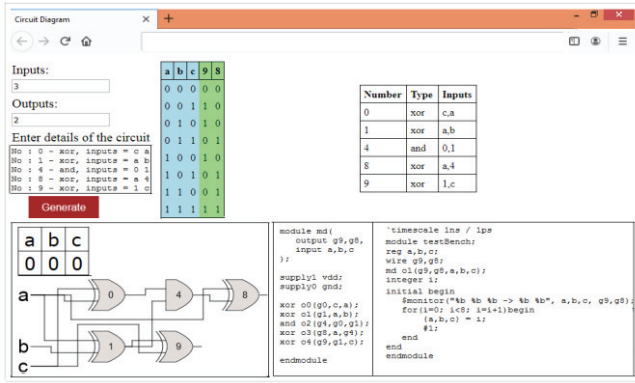


Fig. 4. GUI screen of proposed software

is well known that the gate count is not linearly proportional to transistor count, thus enabling designers to opt for realizing circuits with optimum transistor count. Realization of digital circuits with optimum level count yields minimum propagation delay. The software offers flexibility to designer in terms of selecting the resources required for designing CLCs. This feature has enabled us to compare our design with designs reported in literature using various combinations of resources. Details of various resource combinations used in literature and proposed work are given in Table I, where RT denotes resource type.

TABLE I. RESOURCE TYPES EMPLOYED FOR PROPOSED WORK

#	Resource Type	Resources Used
1	RT1	AND, OR, NOT, XOR
2	RT2	XOR, NAND, NOR, NOT
3	RT3	XOR, XNOR, NAND, NOR, AND, OR, NOT1, NOT2
4	RT4	AND, OR, NAND, NOR, XOR, XNOR, NOT
5	RT5	NOT, NAND, NOR, XOR
6	RT6	2:1 MUX
7	RT7	RM Blocks
8	RT8	AND, OR, XOR, NAND, NOR
9	RT9	AND, OR, XOR, XNOR, NAND, NOR

IV. EXPERIMENTAL RESULTS

The performance evaluation of proposed software is carried out by realizing various CLCs reported in literature and the results are tabulated in Table II. In Table II, D* denotes Boolean expression for a CLC, EAWC denotes evolutionary algorithm without crossover, GC denotes gate count, TC denotes transistor count, GO denotes gate optimization, GS denotes gates saved, TO denotes transistor optimization, TS denotes transistor saved, LO denotes level optimization and SE denotes speed enhancement. The number of levels (L) obtained for the realization of circuits is also reported in Table II. It is observed from Table II that a maximum of 33.33% resources are saved on using proposed work with gate count and transistor count optimization. Transistor count is computed as per the details given in [9], where AND/OR gate is designed using 6 transistors, XOR using 16 transistors, NAND/NOR using 4 transistors and a NOT gate design using 2 transistors. It is also observed that the proposed work enhanced the speed by a factor of 1.5x (max).

The performance evaluation of proposed work is extended towards the realization of CLCs using universal logic modules (2:1 MUX, RM block) and the results are reported in Table III, where RO denotes resource optimization and RS denotes resources saved. It is observed from Table III that a maximum of 33.33% and 25% resources are saved on using proposed work over algorithms reported in literature for 2:1 MUX and RM blocks respectively. It is also observed from Table III that a maximum speed enhancement of 1.50x and 2.00x are obtained on using proposed work over algorithms reported in literature for 2:1 MUX and RM blocks respectively.

In order to assess the performance of proposed work, additional digital circuits (4-bit adder and 2-bit multiplier) are also realized and their results are reported in Table IV. Details of the resources, transistors saved and speed enhancement obtained on using proposed work are reported in Table V. It is observed from Table IV and Table V that 5.00 – 31.25% of resources are saved on using GO over TO and LO. 4.71 – 31.25% of transistors are saved on using TO over GO and LO. Similarly speed enhancement of 1.50x-2.00x is obtained on using LO over GO and TO.

The performance evaluation of proposed work is carried out for few digital benchmark circuits from ISCAS'85 and 74X- series having gate level implementation and the results are reported in Table VI. It may be observed from Table VI that the digital circuits realized using proposed software yielded solutions that are in par or better compared with the resource count reported in benchmark.

V. CONCLUSION

The proposed work was capable of realizing various CLCs reported in literature along with additional circuits like 4-bit adder, 2-bit multiplier and benchmark circuits. Various resource types reported in literature were used to provide a fair comparison between proposed work and others reported in literature. The proposed work provided flexibility to the designer to choose the type of optimization which is employed on a fully functional circuit once the fitness score is satisfied. The proposed work reveals the significance of simulated annealing for design and optimization of combinational logic circuits.

REFERENCES

- [1] Dehui, Z., Faping, Z., Bo, G., et al.: 'Ant Colony Algorithm and Simulated Annealing Algorithm Based Process Route Optimization'. Int. Conf. Enterprise Systems, Shanghai, China, Jan 2015, pp. 1-6
- [2] Ho, L., Shiyu, Y., Wong, C., et al.: 'A simulated annealing algorithm for multiobjective optimizations of electromagnetic devices', IEEE Trans. Magn., 2003,3,(39), pp.1285-1288
- [3] Jai, T., Pratik, D., Rakesh, M.: 'Minimizing Core Supply Noise in a Power Delivery Network by Optimization of Decoupling Capacitors using Simulated Annealing'. IEEE SPI Workshop, Italy, Jun 2017, pp. 1-3.
- [4] Said, K., Hafida, B.: 'Topology-Aware Simulated Annealing'. Int. Conf. Artificial Intelligence, Modelling and Simulation, Spain, May 2015, pp. 1-6
- [5] Wlamir, O., Leonardo, R., Takashi, Y., et al.: 'Troubleshooting optimization using multi-start simulated annealing'. Int. Conf. IEEE Syst. Conf., SYSCON, Orland, FL, USA, Apr 2016, pp.1-6
- [6] Pavitra, Y.J., Jamuna, S., Manikandan, J.: 'Hardware Resource Optimization for Embedded System Design : A Brief Review'. Int. Conf. Convergence in Technology, Pune, India, Apr 2018, pp. 1-6

TABLE II. PERFORMANCE EVALUATION OF PROPOSED WORK FOR VARIOUS CLCS USING VARIOUS RESOURCE TYPES

Ref	Resource Optimized	Algorithm m	Res Type	Literature			Proposed Work (GO)			Proposed Work (TO)			Proposed Work (LO)			
				GC	TC	L	GC	L	%GS	TC	L	%TS	GC	TC	L	SE
D1:F(a,b,c) = Σ(3,5,6)																
[8]	Gate	MLCEA	RT1	4	34	3	4	3	0.00	34	3	0.00	4	34	3	0.00x
[19]	Gate	EAWC	RT1	4	34	3	4	3	0.00	34	3	0.00	4	34	3	0.00x
[9]	Transistor	MLCEA-TC	RT2	4	28	3	4	3	0.00	28	3	0.00	4	28	3	0.00x
[19]	Transistor	EAWC	RT2	4	28	3	4	3	0.00	28	3	0.00	4	28	3	0.00x
[10]	Gate	GA	RT3	4	42	3	4	3	0.00	28	3	33.33	4	28	3	0.00x
[19]	Gate	EAWC	RT3	4	28	3	4	3	0.00	28	3	0.00	4	28	3	0.00x
[13]	Gate	GA	RT9	6	48	3	6	4	0.00	48	4	0.00	6	50	3	0.00x
D2:F(a,b,c,d) = Σ(0,1,3,6,7,8,10,13)																
[8]	Gate	MLCEA	RT1	8	74	5	7	4	12.50	58	4	21.62	7	58	4	1.25x
[19]	Gate	EAWC	RT1	7	58	4	7	4	0.00	58	4	0.00	7	58	4	0.00x
[9]	Transistor	MLCEA-TC	RT2	6	48	4	6	4	0.00	44	4	8.33	6	48	3	1.33x
[19]	Transistor	EAWC	RT2	6	48	4	6	4	0.00	44	4	8.33	6	48	3	1.33x
D3:F(a,b,c,d) = Σ(0,4,5,6,7,8,9,10,13,15)																
[8]	Gate	MLCEA	RT1	7	68	4	6	4	14.28	52	4	23.52	6	52	4	0.00x
[19]	Gate	EAWC	RT1	6	66	4	6	4	0.00	52	4	21.21	6	52	4	0.00x
[9]	Transistor	MLCEA-TC	RT2	7	52	4	5	4	28.57	38	5	26.92	6	46	4	0.00x
[19]	Transistor	EAWC	RT2	7	38	5	5	4	28.57	38	5	0.00	6	46	4	1.25x
[11]	Gate/	ICA	RT4	5	30	4	5	4	0.00	30	4	0.00	6	62	3	1.33x
[19]	Transistor	EAWC	RT4	5	30	4	5	4	0.00	30	4	0.00	6	62	3	1.33x
D4:F1(a,b,c,d) = Σ(7,10,11,13,14,15) F2(a,b,c,d) = Σ(2,3,5,6,8,9,12,15) F3(a,b,c,d) = Σ(1,3,4,6,9,11,12,14)																
[8]	Gate	MLCEA	RT1	7	72	3	7	3	0.00	70	4	0.00	7	72	3	0.00x
[19]	Gate	EAWC	RT1	7	72	3	7	3	0.00	70	4	2.77	7	72	3	0.00x
[9]	Transistor	MLCEA-TC	RT2	9	68	3	8	5	11.11	60	4	8.82	8	68	3	0.00x
[19]	Transistor	EAWC	RT2	8	68	3	8	5	0.00	60	4	8.82	8	68	3	0.00x
[11]	Gate/	ICA	RT4	7	45	3	7	3	0.00	45	3	0.00	7	45	3	0.00x
[19]	Transistor	EAWC	RT4	7	45	3	7	3	0.00	45	3	0.00	7	45	3	0.00x
D5:F(a,b,c) = Σ(3,5,6,7)																
[10]	Gate	GA	RT3	4	24	3	4	3	0.00	18	3	25.00	4	18	3	0.00x
[19]	Gate	EAWC	RT3	4	24	3	4	3	0.00	18	3	25.00	4	18	3	0.00x
D6:F(a,b,c) = Σ(0,3,4,5,6)																
[10]	Gate	GA	RT3	4	42	3	4	3	0.00	28	3	33.33	4	28	3	0.00x
[19]	Gate	EAWC	RT3	4	42	3	4	3	0.00	28	3	33.33	4	28	3	0.00x
D7:F(a,b,c,d) = Σ(2,3,5,6,8,9,12,15)																
[10]	Gate	GA	RT3	3	38	3	3	2	0.00	34	4	10.52	3	36	2	1.50x
[19]	Gate	EAWC	RT3	3	36	3	3	2	0.00	34	4	5.55	3	36	2	1.50x
D8:F(a,b,c,d) = Σ(7,10,11,13,14,15)																
[10]	Gate	GA	RT3	5	30	4	5	3	0.00	22	3	26.67	5	22	3	1.33x
[19]	Gate	EAWC	RT3	5	30	4	5	3	0.00	22	3	26.67	5	22	3	1.33x
D9: F1(a,b,c,d) = Σ(15) F2(a,b,c,d) = Σ(10,11,14) F3(a,b,c,d) = Σ(6,7,9,11,13,14) F4(a,b,c,d) = Σ(5,7,13,15)																
[11]	Gate/	ICA	RT4	7	35	2	7	2	0.00	35	3	0.00	7	35	2	0.00x
[19]	Transistor	EAWC	RT4	7	35	2	7	2	0.00	35	3	0.00	7	35	2	0.00x
D10: F(a,b,c,d) = Σ(0,2,4,6,7,8,9,10,13,14,15)																
[12]	Gate	GA	RT8	6	48	3	5	56	3	42	4	12.50	6	48	3	0.00x
[13]	Gate	GA	RT9	6	36	4	6	40	3	34	4	5.88	6	40	3	1.33x
D11:F(a,b,c) = Σ(1,2,4,7)																
[18]	Gate	GP	RT9	3	--	--	2	2	33.33	32	2	--	2	32	2	--
D12:F(a,b,c) = Σ(0,4,6,7)																
[18]	Gate	GP	RT9	3	--	--	3	2	--	28	2	--	3	28	2	--
D13: F(a,b,c,d) = Σ(1,2,4,7,8,11,13,14)																
[18]	Gate	GP	RT9	3	--	--	3	2	--	48	2	--	3	48	2	--
D14: F(a,b,c,d) = Σ(0,4,5,6,7,8,9,10,13,15)																
[18]	Gate	GP	RT9	5	--	--	5	3	--	44	3	--	5	44	3	--
[13]	Gate	GA	RT9	5	48	4	5	4	0.00	38	5	20.83	6	64	3	1.33x

TABLE III. PERFORMANCE EVALUATION OF PROPOSED SOFTWARE FOR VARIOUS DIGITAL CIRCUITS USING UNIVERSAL MODULE

Ref	Algorithm	RT	Literature		Proposed Work (RO)			Proposed Work (LO)		
			RC	L	RC	L	%RS	RC	L	%SE
D10:F(a,b,c) = Σ(0,4,6,7)										
[14]	GA	2:1MUX	1	1	1	1	0.00	1	1	0.00x
[19]	EAWC	2:1MUX	1	1	1	1	0.00	1	1	0.00x
D11:F(a,b,c) = Σ(1,2,4,7)										
[14]	GA	2:1MUX	3	2	2	2	33.33	2	2	0.00x
[19]	EAWC	2:1MUX	3	2	2	2	33.33	2	2	0.00x
[15]	GA	RM	2	2	2	2	0.00	2	2	0.00x
[19]	EAWC	RM	2	2	2	2	0.00	2	2	0.00x
D12:F(a,b,c,d) = Σ(0,4,6,7,8,12,14,15)										
[14]	GA	2:1MUX	1	1	1	1	0.00	1	1	0.00x
[19]	EAWC	2:1MUX	1	1	1	1	0.00	1	1	0.00x
D13:F(a,b,c,d) = Σ(1,2,4,7,8,11,13,14)										
[14]	GA	2:1MUX	3	3	3	3	0.00	4	2	1.50x
[19]	EAWC	2:1MUX	3	3	3	3	0.00	4	2	1.50x
D14:F(a,b,c,d,e) = Σ(0,1,9,22,23,25,30,31)										
[14]	GA	2:1MUX	5	3	4	3	20.00	4	3	0.00x
[19]	EAWC	2:1MUX	4	3	4	3	0.00	4	3	0.00x
D1:F(a,b,c) = Σ(3,5,6)										
[15]	GA	RM	3	3	3	2	0.00	3	2	0.00x
[19]	EAWC	RM	3	3	3	2	0.00	3	2	0.00x
D15:F(a,b,c,d)= Σ(1,2,3,5,7,8,12)										
[15]	GA	RM	4	4	3	2	25.00	3	2	2.00x
[19]	EAWC	RM	4	4	3	2	25.00	3	2	0.00x
D16:F(a,b,c,d) = Σ(1,2,4,11,13,14)										
[15]	GA	RM	3	3	3	3	0.00	3	2	1.50x
[19]	EAWC	RM	3	3	3	3	0.00	3	2	1.50x

TABLE IV. PERFORMANCE EVALUATION OF PROPOSED SOFTWARE FOR ADDITIONAL DIGITAL CIRCUITS

Circuit	Resource Optimized	Resource Type	Proposed Work (GO)/(RO)			Proposed Work (TO)			Proposed Work (LO)		
			<i>GC/RC</i>	<i>TC</i>	<i>L</i>	<i>GC</i>	<i>TC</i>	<i>L</i>	<i>GC</i>	<i>TC</i>	<i>L</i>
4 bit adder	Gate	RT5	19	170	10	20	162	10	22	204	6
	MUX	RT6	11	220	7	11	220	7	16	320	4
	RM	RT7	13	195	7	13	195	7	14	210	4
2 bit multiplier	Gate	RT5	7	40	2	7	40	2	7	40	2
	MUX	RT6	7	140	3	7	140	3	8	160	2
	RM	RT7	5	75	4	5	75	4	6	90	2

TABLE V. RESOURCES SAVED AND SPEED ENHANCEMENT FOR ADDITIONAL DIGITAL CIRCUITS USING PROPOSED SOFTWARE

Circuit	Resource Type	Resources Saved		Transistors Saved		Speed Enhancement	
		<i>GO over TO</i>	<i>GO over LO</i>	<i>TO over GO</i>	<i>TO over LO</i>	<i>LO over GO</i>	<i>LO over TO</i>
4 bit adder	RT5	5.00%	13.64%	4.71%	20.58%	1.67x	1.67x
	RT6	0.00%	31.25%	0.00%	31.25%	1.75x	1.75x
	RT7	0.00%	7.14%	0.00%	7.14%	1.75x	1.75x
2 bit multiplier	RT5	0.00%	0.00%	0.00%	0.00%	0.00x	0.00x
	RT6	0.00%	12.50%	0.00%	12.50%	1.50x	1.50x
	RT7	0.00%	16.67%	0.00%	16.67%	2.00x	2.00x

TABLE VI. PERFORMANCE EVALUATION OF PROPOSED WORK WITH DIGITAL BENCHMARK CIRCUITS

#Benchmark	Circuit	Resources Used		
ISCAS'85	C17	[20]	6	2 input gate NAND-6
		Proposed Work	6	2 input gate NAND-6
74X-Series	74182	[21]	19	4 input gates (OR-2,AND-4,NOR-1), 3 input gates(NOR-1,AND-1), 2 input gates (AND-5,NOR-1), NOT-1
		Proposed Work	14	4 input gates (OR-3,AND-2), 3 input gates (AND-1,NOR-1), 2 input gates (AND-3,NOR-2), NOT-2
74X-Series	74283	[21]	36	5 input gates (AND-1,NOR-1), 4 input gates (AND-2, OR-1), 3 input gates (AND-3,NOR-1), 2 input gates (AND-4, NAND-4,NOR-5, BubbleAND-4, XOR-4),NOT-2, Buffer-4
		Proposed Work	32	5 input gates (AND-1), 4 input gates (OR-2, AND-2, NOR-4), 3 input gates (AND-5,NOR-2), 2 input gates (AND-4, NOR-2, XOR-9, OR-2)

- [7] Coello,C., Aguirre, A., Buckles,B.: 'Evolutionary multiobjective design of combinational logic circuits'. Int. Conf. Evolvable Hardware, Proc. 2nd NASA/DoD Workshop on Evolvable Hardware, Palo, CA, USA, Aug 2002, pp. 161-170
- [8] Slowik, A., Bialko, M.: 'Design and optimization of combinational digital circuits using modified evolutionary algorithm'. Int. Conf. Artificial Intelligence and Soft Computing, Poland, June 2004, pp. 468-473
- [9] Slowik, A., Bialko, M.: 'Evolutionary design and optimization of combinational digital circuits with respect to transistor count', Bulletin of the POLISH Academy of Sciences Technical Sciences, Vol.54, No.4, 2006
- [10] Chong, K., Aris, L., Bashi, S., et al.: 'Design of Digital Circuit Structure Based on Evolutionary Algorithm Method', Elec. Eng. Tech., 2008, pp. 43-51
- [11] Mehdi, A., Mahani, A., Beig,M.: 'Evolutionary Design and Optimization of Digital Circuits using Imperialist Competitive Algorithm', Comp.Appl., Oct 2011, pp.1-6
- [12] Peeyush, S., Trailokya, N., S. : 'Minimization of Combinational Digital Circuit using Genetic Algorithm'. Int. Conf. CICR, Madurai, India, March 2016, pp. 1-4
- [13] Zahra, A., Ehsan, B. : 'Improving Genetic Algorithm through Clustering for Designing of Combinational Logic Circuits'. Int. Conf. IPRIA, Shahrekod,Iran, July 2017, pp. 1-6
- [14] Vijayakumari, C., Mythili, P., Rekha, K., et al.: 'Genetic Algorithm based design of Combinational Logic Circuits using Universal Logic Modules'. Int. Conf. Information and Communication Technologies, Kochi, India, Dec 2014, pp. 1246-1253
- [15] Vijayakumari, C., Mythili, P., Rekha, K., 'Genetic Algorithm based Design of Combinational Logic Circuits using Reed Muller blocks'. Proc. World Congress on Engineering, London, Jul 2015, pp. 1-6
- [16] Vijayakumari, C., Mythili, P., Rekha, K., et al.: 'Optimal design of Combinational logic Circuits using Genetic Algorithm and Reed Muller Universal Logic modules'. Int. Conf. Embedded Systems, Coimbatore, India, Nov 2014, pp. 1-6
- [17] Karakatic, S.,Podgorelec, V., Hericko, M.: 'Optimization of Combinational Logic Circuits with Genetic Programming', Elec. IR. Elec., 2013, pp. 1-4
- [18] Peeyush, S., Trailokya, N., S. : 'Minimization of Digital Logic Circuit using Genetic Programming with Modified Fitness Function'. Int. Conf. iCATecT, Bangalore, India, Apr 2017, pp.1-5
- [19] Pavitra, Y., Arun, E., Jamuna, S., et al.: 'Study and Evaluation of Digital Circuit Design using Evolutionary Algorithm'. Int. Conf. INDICON-2018, Coimbatore, India, Dec 2018, pp. 1-5
- [20] ISCAS'85 Benchmark circuit
<http://www.pld.ttu.edu/~maksim/benchmarks/>
- [21] Benchmarks-University of Michigan
<http://web.eecs.umich.edu/~jhayes/iscas.restore/benchmark.html>